



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER

BE 11 PHYSICS

TOTAL -60 HOURS

OBJECTIVE:

The students are to be learn the below mentioned:

- Electrostatic in vacuum
- Electrostatic in linear dielectric medium
- Magnetostatic in linear magnetic medium
- Faraday's law
- Electromagnetic waves

Module 1: Wave nature of particles and the Schrodinger equation (12 HOURS)

Introduction to Quantum mechanics, Wave nature of Particles, operators, Time-dependent and time independent Schrodinger equation for wave function, Application: Particle in a one dimensional Box, Born interpretation, Free-particle wave function and wave-packets, v_g and v_p relation Uncertainty principle. Module

2: Wave optics ((12 HOURS)

Huygens' principle, superposition of waves and interference of light by wave front splitting and amplitude splitting: Young double slit experiment, Newton's rings, Michelson interferometer, MachZehnder interferometer, Fraunhofer diffraction from a single slit and a circular aperture, the Rayleigh criterion for limit of resolution and its application to vision; Diffraction gratings and their resolving power. Module

3: Introduction to solids ((12 HOURS)

Free electron theory of metals, Fermi level of Intrinsic and extrinsic, density of states, Bloch's theorem for particles in a periodic potential, Kronig-Penney model (no derivation) and origin of energy bands. V-I characteristics of PN junction, Zener diode, Solar Cell, Hall Effect.

Module 4: Lasers ((12 HOURS)

Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers; gas lasers (He-Ne, CO₂), solid-state lasers (ruby, Neodymium), Properties of laser beams: mono-chromaticity, coherence, directionality and brightness, laser speckles, applications of lasers in science, engineering and medicine. Introduction to Optical fiber, acceptance angle and cone, Numerical aperture, V number, attenuation. Module

5: Electrostatics in vacuum (12 HOURS)

Calculation of electric field and electrostatic potential for a charge distribution; Electric



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

displacement, Basic Introduction to Dielectrics, Gradient, Divergence and curl, Stokes' theorem, Gauss Theorem, Continuity equation for current densities; Maxwell's equation in vacuum and non-conducting medium, Poynting vector.

List of Experiment

1. To determine the dispersive power of prism.
2. To determine the λ of sodium light with the help of newton's Ring.
3. Resolving Power of Telescope.
4. YDSE (Young's double slit Experiment).
5. To determine the frequency of AC mains supply.
6. V-I Characteristics of P-N junction diode.
7. To determine the λ of diode lines by single slit diffraction.
8. To determine the plank's constant with the help of photocell.
9. Hall's effect experiment.
10. Calibration of ammeter by using reference zener diode. w.e.f. July 2018
11. To study the effect of temperature on reverse saturation current in P-N junction diode and to determine the energy band gap.
12. To determine the λ of sodium by using plane diffraction grating.
13. To determine the prominent lines of mercury source by plane diffraction grating.
14. To determine the numerical aperture of an optical fiber.
15. To determine λ of given laser by plane diffraction grating.

OUTCOME:

- Experimental exposure to electromagnetic induction and electromagnetic braking.
- LC circuit and LCR circuit
- Resonance phenomenon in LCR circuit
- Magnetic field from Helmholtz coil
- Measurement of Lorentz force in a vacuum tube

Suggested Reference Books

1. A. Ghatak, Optics.
2. O. Svelto, Principles of Lasers.
3. David Griffiths, Introduction to Electrodynamics.
4. D. J. Griffiths, Quantum Mechanics.
5. Halliday & Resnick, Physics.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER BE 12 MATHEMATICS-I

TOTAL -60 HOURS

OBJECTIVE:

- The objective of their course is to familiarize the prospective engineers with techniques in calculus, multivariate analyzer and linear algebra. it aim to equip the students with standard concepts and tools at an intermediate to advance level that will serve them well towards taking more advanced level of mathematics and applications that they would find useful in their disciplines.

Module 1: Calculus:

Evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. (12 HOURS)

Module 2: Calculus:

Rolle's Theorem, Mean value theorems, Taylor's and Maclaurin theorems with remainders; indeterminate forms and L'Hospital's rule; Maxima and minima. (12 HOURS)

Module 3: Sequences and series:

Convergence of sequence and series, tests for convergence; Power series, Taylor's series, series for exponential, trigonometric and logarithm functions; Fourier series; Half range sine and cosine series, Parseval's theorem. (12 HOURS)

Module 4: Multivariable Calculus (Differentiation):

Limit, continuity and partial derivatives, directional derivatives, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers; Gradient, curl and divergence. (12 HOURS)

Module 5: Matrices

Inverse and rank of a matrix, rank-nullity theorem; System of linear equations; Symmetric, skewsymmetric and orthogonal matrices; Determinants; Eigenvalues and eigenvectors; Diagonalization of matrices; Cayley-Hamilton Theorem, and Orthogonal transformation.

(12 HOURS)

OUTCOME:

The student will learn

- To apply differential and integral calculus to notions of curvature and to improper integrals, apart from some other applications they will have a basic understanding of beta and gamma functions.
- The fallout of Rolle's Theorem that is fundamental to application of analysis to engineering problems.
- The tool of power series and Fourier series for learning advanced Engineering Mathematics.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

- To deal with functions of several variables those are essential in most branches of engineering.
- The essential tool of matrices and linear algebra in comprehensive manner.

BOOKS:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
3. Vecrarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
4. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
5. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
6. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications Reprint, 2008.
7. B.S. Rewal, Higher Engineering Mathematics, Kanna Publishers, 36th Edition, 2010.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER

BE 13 BASIC ELECTRICAL & ELECTRONICS ENGINEERING

TOTAL -60 HOURS

OBJECTIVE

- To understand and analysis basic electrical and magnetic circuits
- To study the working principle of electrical machine and power converters
- To introduce the components of low voltage electrical installations.

MODULE I

Electrical circuit analysis- Voltage and current sources, dependent and independent sources, source conversion, DC circuits analysis using mesh & nodal method, Thevenin's & superposition theorem, star-delta transformation, phase AC circuits under sinusoidal steady state, active, reactive and apparent power, physical meaning of reactive power, power factor, 3-phase balanced and unbalanced supply, star and delta connections. **(12 HOURS)**

Module II

Transformers- Review of laws of electromagnetism, mmf, flux, and their relation, analysis of magnetic circuits, Single-phase transformer, basic concepts and construction features, voltage, current and impedance transformation, equivalent circuits, phasor diagram, voltage regulation, losses and efficiency, OC and SC test. **(12 HOURS)**

Module III

Rotating Electric machines- Constructional details of DC machine, induction machine, and synchronous machine, Working principle of 3-Phase induction motor, Emf equation of 3-Phase induction motor, Concept of slip in 3-Phase induction motor, Explanation of Torque-slip characteristics of 3-Phase induction motor, Classification of self excited DC motor and generator.

(12 HOURS)

Module IV

Digital Electronics- Number systems used in digital electronics, decimal, binary, octal, hexadecimal, their complements, operation and conversion, floating point and signed numbers, Demorgan's theorem, AND, OR, NOT, NOR, NAND, EX-NOR, EX-OR gates and their representation, truth table, half and full adder circuits, R-S flip flop, J-K flip flop. **(12 HOURS)**

Module V

ELECTRONIC COMPONENTS AND CIRCUITS- Introduction to Semiconductors, Diodes, V-I characteristics, Bipolar junction transistors (BJT) and their working, introduction to CC, CB & CE transistor configurations, different configurations and modes of operation of BJT, DC biasing of BJT.

(12 HOURS)



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

GUT COME:

- Get an exposure to common electrical component and their ratings
- Make electrical connections by wire of appropriate ratings.
- Understand the usages of common electrical measuring instruments.
- Understand the basic characteristics of transformer and electrical machines.
- Get an exposure to the working of power electronic converters.

BOOKS:

1. Vincent Del Toro, Electrical Engineering Fundamentals, PHI Learning, II Edition
2. S.Ghosh, Fundamentals of Electrical and Electronics Engineering, PHI, II Edition.
3. Millman, Halkias & Parikh, Integrated Electronics, Mc Graw Hill, II Edition
4. Nagrath & Kothari, Basic Electrical Engineering, III Edition TMH.
5. J.S. Katre, Basic Electronics Engg, Max Pub. Pune.
6. Hughes, Electrical and Electronic Technology, Pearson Education IX Edition



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER

BE 14 ENGINEERING GRAPHICS

TOTAL -60 HOURS

OBJECTIVE:

- All phase of manufacturing or construction requires the conversion of new ideas and design concept into the basic line language of graphics.

Module I

Scales: Representative factor, plain scales, diagonal scales, scale of chords.

Conic sections: Construction of ellipse, parabola, hyperbola by different methods; Normal and Tangent.

Special Curves: Cycloid, Epi-cycloid, Hypo-cycloid, Involute, Archimedean and logarithmic spirals.

(10 HOURS)

Module II

Projection: Types of projection, orthographic projection, first and third angle projection, **Projection of points and lines**, Line inclined to one plane, inclined with both the plane, True Length and True Inclination, Traces of straight lines. (10 HOURS)

Module III

Projection of planes and solids: Projection of Planes like circle and polygons in different positions; Projection of polyhedrons like prisms, pyramids and solids of revolutions like cylinder, cones in different positions. (10 HOURS)

Module IV

Section of Solids: Section of right solids by normal and inclined planes; Intersection of cylinders.
Development of Surfaces: Parallel line and radial - line method for right solids. (10 HOURS)

Module V

Isometric Projections: Isometric scale, Isometric axes, Isometric Projection from orthographic drawing. (10 HOURS)

Module VI

Computer Aided Drafting (CAD): Introduction, benefit, software's basic commands of drafting entities like line, circle, polygon, polyhedron, cylinders; transformations and editing commands like move, rotate, mirror, array; solution of projection problems on CAD. (10 HOURS)



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

OUTCOME:

The student becomes aware of below:

- Introduction to Engineering drawing covering
- Orthographic projections
- Section of solids
- Isometric projection
- Computer aided drafting

BOOKS:

1. Visvesvaraya Tech. University; A Premier on Computer Aided Engg drawing; VTU Belgaum
2. Bhatt N.D.; Engineering Drawing, Charotar
3. Venugopal K.; Engineering Graphics; New Age
4. John KC; Engg. Graphics for Degree; PHI.
5. Gill P.S.; Engineering Drawing; kataria
6. Jeyopoovan T.; Engineering drawing & Graphics Using AutoCAD; Vikas
7. Agrawal and Agrawal; Engineering Drawing; TMH
8. Shrivastava MB and Rana BC; Engg. drawing; Pearson Education
9. Luzadder WJ and Duff JM; Fundamental of Engg Drawing; PHI
10. Jolhe DA Engg. Drawing an Introduction; TMH
11. Narayana K.L.; Engineering Drawing; Scitech



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE 1st SEMESTER

BE 15 BASIC CIVIL ENGINEERING & ENGINEERING MECHANICS

TOTAL -60 HOURS

OBJECTIVE:

- Resolve the forces.
- Find the resultant of given force system
- Find the reaction of beam
- Find the centre of gravity of composite solid.
- Find M.A., V.R., Efficiency and establish law of machine

Module I Building Materials & Construction :

Stones, bricks, cement, lime, timber-types, properties, test & uses, laboratory tests concrete and mortar Materials: Workability, Strength properties of Concrete, Nominal proportion of Concrete preparation of concrete, compaction, curing. Elements of Building Construction, Foundations conventional spread footings, RCC footings, brick masonry walls, plastering and pointing, floors, roofs, Doors, windows, lintels, stair cases types and their suitability. (12 HOURS)

Module II Surveying & Positioning:

Introduction to surveying Instruments – levels, theodolites, plane tables and related devices. Electronic surveying instruments etc. Measurement of distances – conventional and EDM methods, measurement of directions by different methods, measurement of elevations by different methods. Reciprocal leveling. (12 HOURS)

Module –III Mapping & Sensing:

Mapping details and contouring, Profile Cross sectioning and measurement of areas, volumes, application of measurements in quantity computations, Survey stations, Introduction of remote sensing and its applications. (12 HOURS)

Module IV Forces and Equilibrium:

Graphical and Analytical Treatment of Concurrent and non- concurrent Co- planner forces, free Diagram, Force Diagram and Bow's notations; Application of Equilibrium Concepts: Analysis of plane Trusses; Method of joints, Method of Sections. Frictional force in equilibrium problems

(12 HOURS)

Module – V Centre of Gravity and moment of Inertia:

Centroid and Centre of Gravity, Moment Inertia of Area and Mass, Radius of Gyration, Introduction to product of Inertia and Principle Axes. Support Reactions, Shear force and bending moment Diagram for Cantilever & simply supported beam with concentrated, distributed load and Couple. (12 HOURS)



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

OUTCOME:

The students understand the below mention:

- Verification of law of triangle law of forces and Lami's theorem.
- Verification of law of parallelogram law of forces.
- Determination of moment of inertia of flywheel.
- Determination of coefficient of friction between two given materials.
- Determination of forces in the member of jib crane

BOOKS:

1. S. Ramamrutam & R.Narayanan; Basic Civil Engineering, Dhanpat Rai Pub.
2. Prasad I.B., Applied Mechanics, Khanna Publication.
3. Punmia, B.C., Surveying, Standard book depot.
4. Shesha Prakash and Mogaveer; Elements of Civil Engg & Engg. Mechanics; PHI
5. S.P, Timoshenko, Mechanics of structure, East West press Pvt.Ltd.
6. Surveying by Duggal – Tata McGraw Hill New Delhi.
7. Building Construction by S.C. Rangwala- Charotar publications House, Anand.
8. Building Construction by Grucharan Singh- Standard Book House, New Delhi
9. Global Positioning System Principles and application- Gopi, TMH
10. R.C. Hibbler – Engineering Mechanics: Statics & Dynamics.
11. A. Borelli & Schmidt- Engineering Mechanics- statics dynamics, Thomson' Books
12. R.K. Rajput, Engineering Mechanics S.Chand & Co.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE Ist SEMESTER BE 16 INTRODUCTION TO YOGA

TOTAL -60 HOURS

OBJECTIVE:

The students to be made aware of yoga, concept of kriya yoga, bandha and mudras, yoga vashitha, nadi and kanda.

MODULE-1

- Meaning of yoga, Definition, Aim and Objective of yoga.
- Concept of ideal Hut and ideal place and its relevance in modern times.
- Do's and Don't for a yoga practitioner.
- Goal of Hatha yoga and relevance of hatha yoga in contemporary times. (12 HOURS)

MODULE -2

- Concept of kriya yoga.
- Concept of Asanas (Definition, various classification, number of Asanas).
- Concept of pranayama (Definition and types).
- Concept of shhat karma (Dhoti, Neti, Vasti, Kapalbhathi, Tratak, Noli, and Agnisaar). (12 HOURS)

MODULE -3

- Concept, definition of bandha and mudras.
- Concept, definition, benefits and techniques of pratyahar (Patanjalean and Hathyogic).
- Concept of Dharna and dhyana (Patanjali and non Patanjali Concept). (12 HOURS)

MODULE -4

- Highlights of yoga vashitha.
- Yamas, Niyamas, Asanas, Pranayam, Pratyahar, Dharna And Samadhi According to vashitha sanhita.
- Concept of vayu-prana 5 principal pranas function, location, 5 Uppranas sub ordinate function, location.
- Concept of Nadi and Kanda their location and range, its importance in yoga. (12 HOURS)

MODULE -5

- Raja yoga.
- Gyan Yoga.
- Karma Yoga.
- Bhakti Yoga.
- Mantra Yoga.
- Importance of Haths yoga in daily life and social life. (12 HOURS)



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

PRACTICAL

ASANA

1. Halasana
2. Pashchimoutanasana
3. Ardha Matsyendra Asana
4. Akranadhanurasana
5. Uttan Kurmasana
6. Trikorasana
7. Matsyendrasana
8. Kukkutasana
9. Naukasana
10. Garulasana
11. Karnaperasana
12. Padangusthasana
13. Vatayanasana
14. Guptasana
15. Ugrasana
16. Vipreetkarni
17. Tadaasana
18. Sarvangasana
19. Varkshasana

PRANAYAMA

- | | |
|-------------------|--------------|
| 1. Anulom – vilom | 4. Shitali |
| 2. Sitkari | 5. Bhastrika |
| 3. Ujjayi | |

MUDRA

- | | |
|---------------|----------------|
| 1. Yoga Mudra | 2. Brahm Mudra |
|---------------|----------------|

BHANDHA

- | | |
|----------------------|-------------------|
| 1. Jalandhar Bhandha | 2. Udiyan Bhandha |
|----------------------|-------------------|

MEDITATION

1. 10-15 minutes

OUTCOME:

The students having learned types of asana, pranayama, mudra, bhandha and Meditation.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

NATIONAL CADET CORPS



GROOMING TOMORROW'S LEADERS

OFFERING NCC A -GENERAL -GENERIC ELECTIVE CREDIT
COURSE IN UNIVERSITIES UNDER CHOICE BASED CREDIT
SYSTEM TO ALIGN WITH NEW EDUCATION POLICY 2020

CONTENTS

1. Section I . NCC Credit Course Design
2. Section II : NCC Credit Course Rules & Regulations aligned to UGC.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

Training schedules planned for cadets ensure that the optimum benefits of the NCC organization reach maximum number of cadets. The main emphasis is on practical training which in consonance with theory is made to facilitate active participation of learner, better assimilation of knowledge, and proper development of various skills, strengthening of mind and body which is the bedrock of NCC training.

NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	01	01	-	02	
Semester - II	01	01	-	02	
Semester - III	01	01	05	07	Credits merged with 3rd Sem Camp
Semester - IV	02	01	-	03	
Semester - V	01	01	05	07	Credits merged with 5th Sem
Semester - VI	02	01	-	03	
Total	08	06	10	24	Twenty-Four Credits

INSTITUTIONAL TRG SYLLABUS

COMMON SUBJECTS				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	NCC General	06	-	06
2	National Integration	04	-	04
3	Drill	-	45	45
4	Weapon Training	-	25	25
5	Personality Development	25	-	25
6	Leadership	12	-	12
7	Disaster Management	13	-	13
8	Social Service & Community Development	08	39	47
9	Health & Hygiene	-	10	10
10	Adventure	01	-	01
11	Environmental awareness & Conservation	03	-	03
12	Obstacle Training	-	09	09
13	General Awareness	04	-	04
14	Border & Coastal Areas	06	-	06
TOTAL HOURS COMMON SUBJECTS(a)		82	128	210



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER

BE 21 CHEMISTRY-I

TOTAL -60 HOURS

OBJECTIVE:

The concept developed in their course will aid in qualification of several concepts in chemistry that have been introduced at 10+0 levels in schools. Technology is being increasingly based on the electronic, atomic and molecular level modifications. Quantum theory is more than 100 years old and to understand phenomenon at nanometer levels, one has to be base the description of all chemical processes at molecular levels.

Course Contents:

MODULE -I

Water – Analysis, Treatments and Industrial Applications

Sources, Impurities, Hardness & its units, Determination of hardness by EDTA method, Alkalinity & It's determination and related numerical problems. (9 hours)

MODULE -II

Boiler problem & softening methods

Boiler troubles (Sludge & Scale, Priming & Foaming, Boiler Corrosion, Caustic Embrittlement), Softening methods (Lime-Soda, Zeolite and Ion Exchange Methods) and related numerical problems. (9 hours)

MODULE -III

Lubricants and Lubrication

Introduction, Mechanism of lubrication, Classification of lubricants, significance & determination of Viscosity and Viscosity Index, Flash & Fire Points, Cloud & Pour Points, Aniline Point, Acid Number, Saponification Number, Steam Emulsification Number and related numerical problems. (9 hours)

MODULE -IV

Polymer & polymerization

Introduction, types of polymerisation, Classification, mechanism of polymerisation (Free radical & Ionic polymerization). Thermoplastic & Thermosetting polymers Elementary idea of Biodegradable polymers, preparation, properties & uses of the following polymers- PVC, PMMA, Teflon, Nylon 6, Nylon 6:6, Polyester phenol formaldehyde, Urea- Formaldehyde, Buna N, Buna S, Vulcanization of Rubber. (9 hours)

MODULE -V

Phase equilibrium and Corrosion

Phase diagram of single component system (Water) Phase diagram of binary Eutectic System (Cu-Ag.) Corrosion: Types, Mechanisms & prevention. (9 hours)

MODULE -VI

Spectroscopic techniques and application



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

Ammonium Sulphate & Copper Sulphate.

Reference Books :

- 1 Chemistry in Engineering and Technology - Vol.1 &2 Kuriacose and Rajaram , McGraw Hill Education
- 2 Fundamental of Molecular Spectroscopy C.N. Banwell , McGraw Hill Education
- 3 Engineering Chemistry – B.K. Sharma, Krishna Prakashan Media (P) Ltd., Meerut.
- 4 Basics of Engineering Chemistry – S.S. Dara & A.K. Singh, S. Chand & Company Ltd., Delhi.
- 5 Applied Chemistry – Theory and Practice, O.P. Viramani, A.K. Narula, New Age International Pvt. Ltd. Publishers, New Delhi.
- 6 Elementary Spectroscopy , Y.R. Sharma , S. Chand Publishing
- 7 Polymer Science, Vasant R. Gowariker, N. V. Viswanathan, Jayadev Sreedhar, New Age International Pvt. Ltd
- 8 Advanced Inorganic Chemistry, G.R. Chatwal, Goal Publishing House
- 9 Engineering Chemistry (NPTEL Web-book) B.L. Tembe, Kamaluddin and M.S. Krishna



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 22 MATHEMATICS-II

OBJECTIVE:

TOTAL -60 HOURS

- The objective of their course is to familiarize the prospective engineers with techniques in multivariable integration ordinary and partial differential equations and complex variables. It aims to equip the students to deal with advance level of mathematics and applications that would be essential for their disciplines.

Module 1

Ordinary Differential Equations I : (6 hours) : Differential Equations of First Order and First Degree (Leibnitz linear, Bernoulli's, Exact), Differential Equations of First Order and Higher Degree, Higher order differential equations with constants coefficients, Homogeneous Linear Differential equations, Simultaneous Differential Equations. (12 HOURS)

Module 2

Ordinary differential Equations II: (8 hours) : Second order linear differential equations with variable coefficients, Method of variation of parameters, Power series solutions; Legendre polynomials, Bessel functions of the first kind and their properties. (12 HOURS)

Module 3:

Partial Differential Equations : (8 hours) : Formulation of Partial Differential equations, Linear and Non-Linear Partial Differential Equations, Homogeneous Linear Partial Differential Equations with Constants Coefficients. (12 HOURS)

Module 4

Functions of Complex Variable : (8 hours) : Functions of Complex Variables: Analytic Functions, Harmonic Conjugate, Cauchy-Riemann Equations (without proof), Line Integral, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Singular Points, Poles & Residues, Residue Theorem, Application of Residues theorem for Evaluation of Real Integral (Unit Circle).

(12 HOURS)

Module 5

Vector Calculus : (10 hours) : Differentiation of Vectors, Scalar and vector point function, Gradient, Geometrical meaning of gradient, Directional Derivative, Divergence and Curl, Line Integral, Surface Integral and Volume Integral, Gauss Divergence, Stokes and Green theorems. (12 HOURS)

OUTCOME:

The student will learn

- The mathematical tools needed in evaluating multiple integrals and their usages.
- The effective mathematical tools for the solutions of differential equations that model physical



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

processes.

- The tools of differentiation and integration of functions of a complex variable that are use various techniques dealing engineering problem.

Textbooks/References:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
3. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edn., Wiley India, 2009.
4. S. L. Ross, Differential Equations, 3rd Ed., Wiley India, 1984.
5. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India, 1995.
6. E. L. Ince, Ordinary Differential Equations, Dover Publications, 1958.
7. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Ed., McGraw Hill, 2004.
8. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
9. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER

BE 23 PROGRAMMING FOR PROBLEM SOLVING WITH C++

TOTAL -60 HOURS

OBJECTIVE :

- To formulate simple algorithms for erythematic and logical problem
- To translate the algorithms to programs (in c++ language)
- To test and execute the programs and correct syntax and logical error.
- To implement conditional branching, iteration recursion.
- To decompose a problem into functions and synthesize a complete program using divide and Conquer approach.
- To use arrays, pointer and structures to formulate algorithms and program.
- To apply programming to solve matrix addition and multiplication problems and searching and sorting problem.

Module -I

Introduction To 'C' Language

Character set, variables and identifiers, built-in data types, variable definition, arithmetic operators and expressions, constants and literals, simple assignment statement, basic input/output statement, simple 'c' programs. (09 HOURS)

Module -II

Conditional Statements And Loops

Decision making within a program, conditions, relational operators, logical connectives, if statement, if-else statement, loops: while loop, do while, for loop, nested loops, infinite loops, switch statement, structured programming. (09 HOURS)

Module -III Arrays

One dimensional arrays: array manipulation; searching, insertion, deletion of an element from an array; finding the largest/smallest element in an array; two dimensional arrays, addition/multiplication of two matrices, transpose of a square matrix; null terminated strings as array of characters, representation sparse matrices. (09 HOURS)

Module -IV

Functions

Top-down approach of problem solving, modular programming and functions, standard library of c



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

functions, prototype of a function: formal parameter list, return type, function call, block structure, passing arguments to a function: call by reference, call by value, recursive functions, arrays as function arguments. **(09 HOURS)**

Module –V

Structures and Unions

Structure variables, initialization, structure assignment, nested structure, structures and functions, structures and arrays: arrays of structures, structures containing arrays, unions. **(08 HOURS)**

Module –VI

Pointers

Address operators, pointer type declaration, pointer assignment, pointer initialization, pointer arithmetic, functions and pointers, arrays and pointers, pointer arrays. **(08 HOURS)**

Module –VII File Processing

Concept of files, file opening in various modes and closing of a file, reading from a file, writing onto a file. **(08 HOURS)**

OUTCOME:

- To formulate the algorithms for simple problems.
- To translate given algorithms to a working and correct program.
- To be able to correct syntax errors or reported by the compiler.
- To be able to identify and correct logical error encountered at run time.
- To be able to write iterative as well as recursive programs.

Reference

1. Let Us C-Yaswant kanetkar, BPP
2. Programming in ANSI C-E. Balagurusamy, TMH



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 24 ENGLISH

TOTAL -60 HOURS

OBJECTIVE:

The student will acquire basic proficiency in English including reading and listening comprehension, writing and speaking skill.

Module I - Languages and skills of communication

Linguistic techniques, Modern usages, Reading comprehension, English phonetic symbols/sounds, Oral presentation, Audition Communication, Processes of Communication, Verbal and Non Verbal Communication, Barriers to Communication. (12 HOURS)

Module II - Application of linguistic ability

Writing of definitions of Engineering terms, Objects, Processes and Principles (Listening) Topics of General Interest, Reproduction from business, daily life, travel, health, buying and selling, company structure, systems etc. (12 HOURS)

Module III - Letter Writing:

Applications, Enquiry, Calling quotations, Tenders, Order and Complaint. (12 HOURS)

Module IV

Precise Writing, Noting and drafting, Technical Description of simple engineering objects and processes (writing), Report writing, precise writing, Note writing, Slogan writing comment, Speech advertising. (12 HOURS)

Module V

Writing Technical reports of the type of observation report, Survey report, Report of trouble, Laboratory Report and Project Report on the subjects of engineering. (Speaking) Vocabulary, Presentations, Demonstrations, Conversation- Telephone media, socializing, cultural events, debates, speech. (12 HOURS)

OUTCOME:

The student understands the below mention:

- Language and skills of communication
- Application linguistic ability
- Letter writing precise writing
- Writing technical reports



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

- To be able to represent data in arrays, strings and structure and manipulate them through a program.
- To be able to declare pointers of different types and use them in defining self-referential structures.
- To be able to create, read and write to and from simple text files.

Topics to be covered in the Language laboratory sessions

1. Basic Grammar & Vocabulary (Synonyms /Antonyms, Analogies, sentence completion, correctly spelt words, idioms, proverbs, common errors).
2. Phonetic symbols and pronunciation.
3. Listening skills (Including Listening Comprehension)
4. Reading Skills (Including Reading Comprehension)
5. Writing Skills (Including structuring resume and cover letter)
6. Speaking Skills
7. Body Language
8. Oral Presentation: Preparation and delivery using audio – visual aids with stress n body language And voice modulation (Topic to be selected by the teacher.) Final Assessment Should be based on Assignment, presentation and interview.

BOOKS:

1. Business Correspondence and Report Writing - By Sharma; TMH.
2. Living English Structure – By W.S. Allen; Longmans.
3. English Grammar – Ehrlich, Schaum Series; TMH.
4. Spoken English for India – By R.K. Bansal and IB Harrison Orient Longman.
5. New International Business English – by Joans and Alexander; OUP.
6. Effective Technical Communication – Rizvi; TMH.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 25 ENVIROMENTAL STUDIES

TOTAL -60 HOURS

Module –I Energy-

Sources of Energy : Renewable & Non Renewable, Fossil fuel, Biomass Geothermal, Hydrogen, Solar, Wind, hydal, nuclear sources. (12 HOURS)

Module –II Ecosystem –

Segments of Environment: Atmosphere, hydrosphere, Lithosphere, biosphere. Cycles in Ecosystem – Water, Carbon, Nitrogen. Biodiversity: Threats and conservation, (12 HOURS)

Module –III Air Pollution & Sound Pollution

Air Pollution: Air pollutants, classification, (Primary & secondary Pollutants) Adverse effects of pollutants. Causes of Air pollution chemical, photochemical, Green house effect, ozone layer depletion, acid Rain. Sound Pollution: Causes, controlling measures, measurement of sound pollution (deciblage), Industrial and non – industrial. (12 HOURS)

Module –IV Water Pollution-

Water Pollution: Pollutants in water, adverse effects. Treatment of Domestic & Industrial water effluent.

Soil Pollution – Soil Profile, Pollutants in soil, their adverse effects, controlling measures.

(12 HOURS)

Module –V Society, Ethics & Human values

Impact of waste on society. Solid waste management (Nuclear, Thermal, Plastic, medical, Agriculture, domestic and e-waste). Ethics and moral values, ethical situations, objectives of ethics and its study Preliminary studies regarding Environmental Protection Acts , introduction to value education, self exploration, sanyam & swasthya. (12 HOURS)

BOOKS:

1. Harris, CE, Prichard MS, Rabin's MJ, "Engineering Ethics"; Cengage Pub.
2. RanaSVS; "EssentialsofEcologyand Environment"; PHIPub.
3. Raynold, GW "Ethics in information Technology"; Cengage.
4. Svakumar; Energy Environment & Ethics in society; TMH
5. AK De "Environmental Chemistry"; New Age Int. Publ.



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

6. BK Sharma, "Environmental Chemistry"; Goel Publ. House.
7. Bala Krishnamoorthy; "Environmental management"; PHI
8. Gerard Kiely, "Environmental Engineering"; TMH
9. Miller GT JR; living in the Environment Thomson/cengage
10. Cunningham WP and MA; principles of Environment Sc; TMH
11. Pandey, S.N. & Mishra, S.P. Environment & Ecology, 2011, Ane Books , Pvt. Ltd, New Delhi
12. Joseph, B. Environmental Studies, 2009 Tata Mcgraw Hill, Edu India Ltd. New Delhi.
13. Gour R.R, Sangal, R & Bagaria, G.P. , Excel Books, A-45, Naraina Phase-I
New Delhi.-110028



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER

BE 26 WORK SHOP/ MANUFACTURING PRACTICE

TOTAL -60 HOURS

OBJECTIVE:

- The student will gain knowledge of the different manufacturing processes which are commonly employed in the industry, to fabricate components using different materials.

Module I

Introduction: Manufacturing Processes and its Classification, Casting, Machining, Plastic deformation and metal forming, Joining Processes, Heat treatment process, Assembly process. Powder Metallurgy: introduction to computers in manufacturing, Black Smithy Shop Use of various smithy tools. Forging operations: Upsetting, Drawing down, Fullering, Swaging, Cutting down, Forge welding, Punching and drafting. Suggested Jobs : Forging of chisel., forging of Screw Driver.

(12 HOURS)

Module II

Carpentry Shop: Timber : Type, Qualities of timber disease, Timber grains, Structure of timber, Timber, Timber seasoning, Timber preservation .Wood Working tools: Wood working machinery, joints & joinery. Various operations of planning using various carpentry planes sawing & marking of various carpentry joints. Suggested Jobs :Name Plate ,Any of the Carpentry joint like mortise or tennon joint. (12 HOURS)

Module III

Fitting Shop: Study and use of Measuring instruments, Engineer steel rule, Surface gauges caliper, Height gauges, feeler gauges, micro meter. Different types of files, File cuts, File grades, Use of surface plate, Surface gauges drilling tapping Fitting operations: Chipping filling, Drilling and tapping.Suggested Jobs :Preparation of job piece by making use of filling, sawing and chipping , drilling and tapping operations. (12 HOURS)

Module IV

Foundry: Pattern Making: Study of Pattern materials, pattern allowances and types of patterns. Core box and core print, Use and care of tools used for making wooden patterns.

Moulding: Properties of good mould & Core sand, Composition of Green , Dry and Loam sand. Methods used to prepare simple green and bench and pit mould dry sand bench mould using single piece and split patterns. (12 HOURS)

Module V

Welding: Study and use of tools used for Brazing, Soldering, Gas & Arc elding. Preparing Lap &



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

Butt joints using gas and arc welding methods, Study of TIG & MIG welding processes. Safety precautions. (12HOURS)

OUTCOME:

- Upon completion of the course, students will be able to fabricate components with their own hands.

BOOKS:

1. Bawa HS; Workshop Practice, TMH
2. Rao PN; Manufacturing Technology- Vol.1 & 2, TMH
3. John KC; Mechanical workshop practice; PHI
4. Hazare Choudhary; Workshop Practices -, Vol. I & II
5. Jain. R.K. Production Technology -



SARVEPALLI RADHAKRISHNAN UNIVERSITY, BHOPAL

BE II nd SEMESTER BE 27 Constitution of India

Basic features and fundamental principles

TOTAL -60 HOURS

The Constitution of India is the supreme law of India. Parliament of India cannot make any law which violates the Fundamental Rights enumerated under the Part III of the Constitution. The Parliament of India has been empowered to amend the Constitution under Article 368, however, it cannot use this power to change the "basic structure" of the constitution, which has been ruled and explained by the Supreme Court of India in its historical judgments. The Constitution of India reflects the idea of "Constitutionalism" – a modern and progressive concept historically developed by the thinkers of "liberalism" – an ideology which has been recognized as one of the most popular political ideology and result of historical struggles against arbitrary use of sovereign power by state. The historic revolutions in France, England, America and particularly European Renaissance and Reformation movement have resulted into progressive legal reforms in the form of "constitutionalism" in many countries. The Constitution of India was made by borrowing models and principles from many countries including United Kingdom and America. (20 HOURS)

The Constitution of India is not only a legal document but it also reflects social, political and economic perspectives of the Indian Society. It reflects India's legacy of "diversity". It has been said that Indian constitution reflects ideals of its freedom movement; however, few critics have argued that it does not truly incorporate our own ancient legal heritage and cultural values. No law can be "static" and therefore the Constitution of India has also been amended more than one hundred times. These amendments reflect political, social and economic developments since the year 1950. The Indian judiciary and particularly the Supreme Court of India have played an historic role as the guardian of people. It has been protecting not only basic ideals of the Constitution but also strengthened the same through progressive interpretations of the text of the Constitution. The judicial activism of the Supreme Court of India and its historic contributions has been recognized throughout the world and it gradually made it "as one of the strongest court in the world".

Historic revolutions in france england america and particularly european renaissance and india and its historic contributions has been recognized throughout the world and it gradually made it "as one of the strongest court in the world". reformation movement have resulted into progressive legal reform in the form of "constitutionalism" in many countries. the constitution of india was made by borrowing models and principles from many countries including united kingdom and america. (20 HOURS)

The Constitution of India is not only a legal document but it also reflects social, political and economic perspectives of the Indian Society. It reflects India's legacy of "diversity". It has been said that Indian constitution reflects ideals of its freedom movement, however, few critics have argued that it does not truly incorporate our own ancient legal heritage and cultural values. No law can be "static" and therefore the Constitution of India has also been amended more than one hundred times. These amendments reflect political,

social and economic developments since the year 1950. The Indian judiciary and particularly the Supreme Court of India has played an historic role as the guardian of people. It has been protecting not only basic ideals of the Constitution but also strengthened the same through progressive interpretations of the text of the Constitution. The judicial activism of the Supreme Court of Course content (20 HOURS)

1. Meaning of the constitution law and constitutionalism
2. Historical perspective of the Constitution of India
3. Salient features and characteristics of the Constitution of India
4. Scheme of the fundamental rights
5. The scheme of the Fundamental Duties and its legal status
6. The Directive Principles of State Policy – Its importance and implementation
7. Federal structure and distribution of legislative and financial powers between the Union and the States
8. Parliamentary Form of Government in India – The constitution powers and status of the President of India
9. Amendment of the Constitutional Powers and Procedure
10. The historical perspectives of the constitutional amendments in India
11. Emergency Provisions : National Emergency, President Rule, Financial Emergency
12. Local Self Government – Constitutional Scheme in India
13. Scheme of the Fundamental Right to Equality
14. Scheme of the Fundamental Right to certain Freedom under Article 19
15. Scope of the Right to Life and Personal Liberty under Article 21.

NATIONAL CADET CORPS



GROOMING TOMORROW'S LEADERS

**OFFERING NCC A -GENERAL -GENERIC ELECTIVE CREDIT
COURSE IN UNIVERSITIES UNDER CHOICE BASED CREDIT
SYSTEM TO ALIGN WITH NEW EDUCATION POLICY 2020**

CONTENTS

3. Section I : NCC Credit Course Design
4. Section II: NCC Credit Course Rules & Regulations aligned to UGC.

SECTION I: NCC CREDIT COURSE DESIGN DOCUMENT

UNDER CHOICE BASED CREDIT SYSTEM AS GENERAL ELECTIVE FOR SENIOR DIVISION / SENIOR WING

4. **Preamble.** The National Cadet Corps (NCC) is governed by NCC Act 1948 and attendant NCC Rules. It functions under the Ministry of Defence and is headed by DGNCC. It is organised into 17 State Directorates each headed by an Additional/Deputy Director General. The aims of NCC are:-

(a) To develop character, camaraderie, discipline, secular outlook, the spirit of adventure, sportsman spirit and ideals of selfless service amongst cadets by working in teams, honing qualities such as self-discipline, self-confidence, self-reliance and dignity of labour in the cadets.

(b) To create a pool of organized, trained and motivated youth with leadership qualities in all walks of life, who will serve the Nation regard less of which career they choose.

(c) To provide a conducive environment to motivate young Indians to choose the Armed Forces as a career.

5. **Purpose.** Currently NCC training is imparted as extra-curricular activity to volunteer students from recognized schools and colleges who enroll as cadets. NCC as a Credit Course is designed with an intent to transform NCC training into a curricular activity from an extra-curricular thereby providing academic credits to students undergoing NCC training along with other attended advantages to the cadets in the college/ university.

6. **Introduction to NCC Credit Course Design.** Institutional Training is the mainstay of NCC training and it is conducted at colleges and universities by Associate NCC Officers and Armed Forces personnel. The application of knowledge gained through institutional training is further honed or developed to a higher degree in NCC Camps. The Institutional Training syllabus comprises Common Subjects and Specialised Subjects

(military component). NCC Credit Course is designed to offer Institutional Training of Senior Wing /Division is over six semesters (three years), comprising 300330 periods (excluding Camp), of which 120150 periods are meant for theory with 1008 credits and 180 periods for practical with 6 credits. Each period is counted as hour. The ratio between theory and practical in terms of number of hours of training is 5:6, but in terms of credits is 5:3, since as per CBCS two hours of practical is counted towards one period of training as against one hour for theory. In addition two separate courses have been designed for two Camps normally referred to as Annual Training Camps (ATC).

Training schedules planned for cadets ensure that the optimum benefits of the NCC organization reach maximum number of cadets. The main emphasis is on practical training which in consonance with theory is made to facilitate active participation of learner, better assimilation of knowledge, and proper development of various skills, strengthening of mind and body which is the bedrock of NCC training.

NCC GENERAL ELECTIVE CREDIT COURSE DESIGN SUMMARY					
Semester	Credits Allocated			Total	Remarks
	Theory	Practical	Camp		
Semester - I	01	01	-	02	
Semester - II	01	01	-	02	
Semester - III	01	01	05	07	Credits merged with 3of rd1 st Sem Camp
Semester - IV	02	01	-	03	
Semester - V	01	01	05	07	Credits merged with 5of
Semester - VI	02	01	-	03	
Total	08	06	10	24	Twenty-Four Credits

INSTITUTIONAL TRG SYLLABUS

COMMON SUBJECTS				
Ser	Subject	Periods (1 hour duration each)		Total
		Lectures/Tutorials	Practicals	
1	NCC General	06	-	06
2	National Integration	04		04
3	Drill	-	45	45
4	Weapon Training	-	25	25
5	Personality Development	25		25
6	Leadership	12	-	12
7	Disaster Management	13		13
8	Social Service & Community Development	08	39	47
9	Health & Hygiene	-	10	10
10	Adventure	01		01
11	Environmental awareness & conservation	03		03
12	Obstacle Training	-	09	09
13	General Awareness	04		04
14	Border & Coastal Areas	06		06
TOTAL HOURS COMMON SUBJECTS(a)		82	128	210

SPECIALISED SUBJECTS (ARMY)